

AI-Enhanced Multisensory Navigation for Visually Impaired Users in Virtual Environments

Luiza M. F. Cintra *

Federal University of Goiás, AKCIT

Andressa A. Bastos

Federal University of Goiás, AKCIT

Rafael T. Sousa

Federal University of Mato Grosso, AKCIT

Elisa A. M. Oliveira

Federal University of Goiás, AKCIT

Valdemar V. G. Neto

Federal University of Goiás

Arlindo R. Galvão Filho †

Federal University of Goiás, AKCIT

Gustavo H. W. Barbosa

AKCIT

Sofia L. C. Paiva

Federal University of Goiás

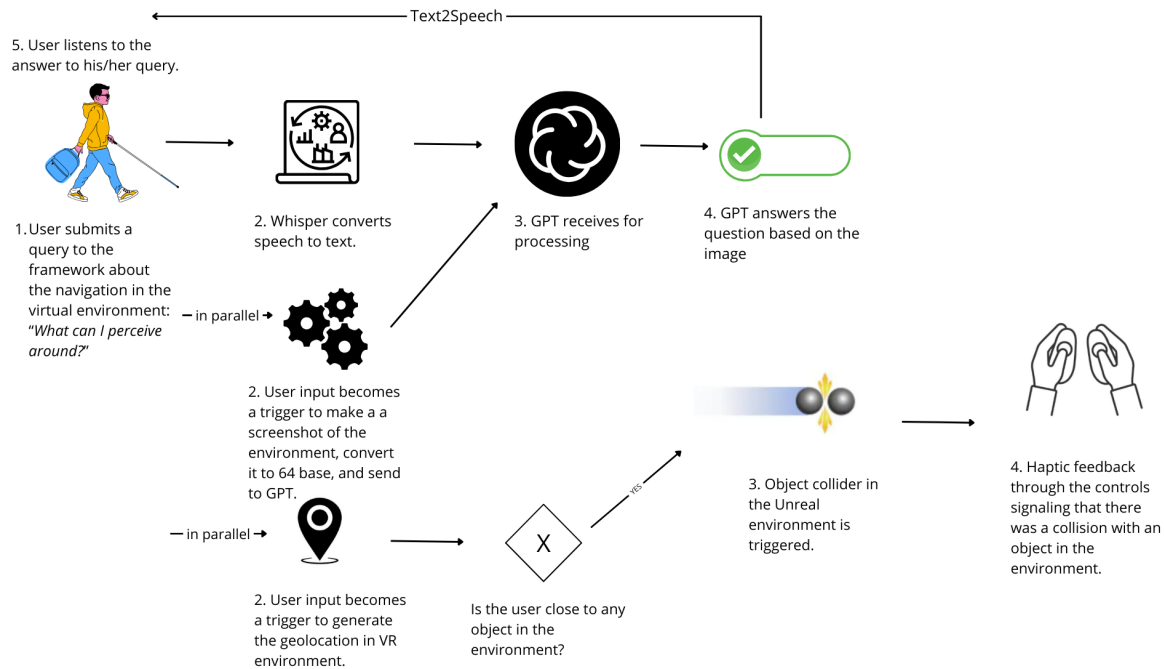


Figure 1: System architecture for AI-based interactive assistance integrated with Unreal Engine. The diagram illustrates the workflow of the proposed system. User input is processed by Whisper-Tiny for transcription and combined with a scene screenshot and geolocation data from Unreal. These inputs are sent to GPT-4o, which interprets the user's intent. The output is used in two ways: generating speech feedback via a text-to-speech module, and triggering object collisions in the virtual environment to provide force feedback and additional audio cues.

ABSTRACT

VR applications often exclude individuals with visual impairments, creating a significant accessibility gap. We present a real-time assistive framework integrating Vision Large Language Models (VLLMs) to interpret complex visual scenes. The system generates context-aware audio descriptions combined with haptic collision feedback, translating visual data into perceptible patterns. A user study with blind (N=3) and low-vision (N=2) participants was conducted in simulated clinic and classroom environments. Findings show that AI-driven assistance significantly enhanced spatial awareness, independence, and task performance. Qualitative results confirmed reduced cognitive load and heightened presence,

providing a robust, VLLM-powered path for designing inclusive immersive systems.

Index Terms: Virtual Reality, Accessibility, Visual Impairment, Artificial Intelligence, Assistive Technology, User Study, Non-Visual Feedback.

1 CORE CONCEPT

The rapid growth of virtual and immersive environments has expanded their use in education, work, and social interaction [1]. However, most VR systems remain predominantly visually oriented, creating significant accessibility barriers for people with visual impairments [3]. This reliance on visual cues often leads to exclusion, reduced autonomy, and higher cognitive effort for blind and low-vision users, particularly in tasks involving spatial navigation and environmental understanding [4, 2, 5].

A central challenge is translating complex visual information—such as spatial layout, object positions, and obstacles—into perceptible and actionable forms without overloading users cognitively. Existing VR accessibility solutions mainly use static or pre-

*e-mail: cintraluiza@discente.ufg.br

†e-mail: arlindogalvao@ufg.br

scripted audio cues, which lack adaptability to dynamic environments and provide limited support for real-time decision-making, resulting in fragmented user experiences.

To overcome these limitations, this work proposes a real-time assistive framework based on multimodal artificial intelligence for context-aware navigation in virtual environments (Figure 1). The system employs Vision Large Language Models (VLLMs) to interpret virtual scenes by combining visual data, spatial metadata, and spoken user input. It delivers adaptive auditory guidance aligned with user intent and complements it with haptic feedback through controlled collisions, transforming visual information into meaningful sensory cues that support spatial understanding, confidence, and independent navigation.

2 RESULTS OF FINDINGS

To demonstrate the feasibility and generalizability of the framework, two distinct immersive VR scenarios were designed to simulate ecologically valid environments: a classroom and a hospital reception hall. The rationale for selecting heterogeneous contexts was to ensure that the framework could support non-visual navigation across structurally different spatial layouts and interaction demands.

Upon arrival, participants were informed about the study and signed an informed consent form (ICF). All data were treated confidentially and anonymously, in accordance with ethical guidelines approved by the Research Ethics Committee (Protocol 88442725.6.0000.5083; 25.07.2025). Following the practice session, participants performed the experimental tasks in virtual environments designed to assess accessibility, immersion, and usability. During navigation, both objective measures (e.g., task completion time, number of collisions, interruptions due to discomfort) and subjective measures (e.g., sense of presence, clarity of instructions, ease of navigation) were collected. After completing each experimental condition, participants were allowed to take short breaks if needed.

The user study involved seven participants (three with total blindness and four with partial vision). Among them, five participants completed all tasks, while two did not complete the full protocol due to difficulties related to VR hardware interaction and task comprehension. The results reported below focus on the five participants who completed all missions, which included three blind users and two users with low vision.

2.1 Task Performance

All five participants successfully completed the full set of navigation tasks in both virtual environments relying exclusively on auditory and haptic feedback. Task performance was primarily assessed using task completion time, measured in seconds for each task. Results show that participants with total blindness required longer completion times than participants with low vision across all tasks. However, observational analysis indicated that this difference decreased over repeated trials, suggesting rapid adaptation to the interaction mechanics and feedback strategy.

Overall task completion times demonstrate that, despite initial differences, both blind and low-vision participants were able to reach comparable performance levels by the end of the session, completing goal-oriented navigation tasks independently within the virtual environments.

2.2 Subjective Measures

In addition to objective performance data, the evaluation incorporated subjective measures collected through a post-study questionnaire and semi-structured interviews. Subjective metrics assessed usability, accessibility and inclusivity, emotional and cognitive impact, system functionality, and multisensory engagement.

Participants consistently reported positive experiences regarding ease of navigation, clarity of auditory instructions, and usefulness of haptic feedback. Feedback indicated that the multimodal guidance reduced uncertainty during navigation, supported spatial understanding, and increased confidence while performing tasks.

While overall perceptions were favorable, participants highlighted accessibility and inclusivity as an area for improvement, noting differences in individual needs related to vision level, familiarity with assistive technologies, and interaction with VR hardware.

2.3 Summary of Findings

Overall, the results show that the proposed framework enables blind and low-vision users to effectively perform navigation tasks in immersive virtual environments. Objective performance metrics indicate successful task completion and user adaptation over time, while subjective evaluations report increased usability, confidence, and multisensory engagement. These findings demonstrate the feasibility of AI-driven multisensory assistance for improving VR accessibility and highlight the need to further address onboarding, hardware usability, and inclusivity in future developments.

3 DISCUSSION: CONTRIBUTION TO INTERACTION AND INTERFACES TOPIC

This work contributes to Interaction and Interfaces by showing how AI-driven semantic interpretation can enable inclusive interaction in immersive 3D environments. The proposed framework integrates multimodal feedback—speech, spatial audio, and haptics—into an adaptive loop that responds in real time to user intent and environmental context.

From a 3D Interaction perspective, the system supports exploratory and goal-oriented navigation through context-aware guidance rather than fixed paths, fostering spatial understanding and user autonomy. In terms of Multimodal Interaction, the coordinated use of auditory and haptic feedback provides semantically aligned, redundant cues that reduce reliance on vision and improve accessibility for visually impaired users.

Overall, the work presents a scalable design approach that embeds AI reasoning into interface logic, with broader implications for adaptive and accessible interaction beyond visually driven systems.

ACKNOWLEDGMENTS

This work has been partially funded by the project Research and Development of Algorithms for Construction of Digital Human Technological Components supported by Advanced Knowledge Center in Immersive Technologies (AKCIT), with financial resources from the PPI IoT/Manufatura 4.0 / PPI HardwareBR of the MCTI grant number 057/2023, signed with EMBRAPPI.

REFERENCES

- [1] F. E. V. d. Anjos, A. d. O. Martins, G. S. Rodrigues, M. A. Sellitto, and D. O. d. Silva. Boosting engineering education with virtual reality: An experiment to enhance student knowledge retention. *Applied System Innovation*, 7(3):50, 2024. 1
- [2] F. Heilemann, G. Zimmermann, and P. Münster. Accessibility guidelines for vr games—a comparison and synthesis of a comprehensive set. *Frontiers in Virtual Reality*, 2:697504, 2021. 1
- [3] S. J. Leat, G. E. Legge, and M. A. Bullimore. What is low vision? a re-evaluation of definitions. *Optometry and Vision Science*, 76(4):198–211, 1999. 1
- [4] D. R. Pur, N. Lee-Wing, and M. D. Bona. The use of augmented reality and virtual reality for visual field expansion and visual acuity improvement in low vision rehabilitation: a systematic review. *Graefes' Archive for Clinical and Experimental Ophthalmology*, 261(6):1743–1755, 2023. 1
- [5] B. Yuan, E. Folmer, and F. C. Harris Jr. Game accessibility: a survey. *Universal Access in the information Society*, 10(1):81–100, 2011. 1